Synaptic Plasticity as Short-Term Memory

One-page concept summary • DataForge 2026 • Explain the Frontier

Core claim. Short-term synaptic plasticity can store recent information in changing synaptic states, but persistence of those traces creates a trade-off between retaining recent information and being influenced by newer or competing information. Our artifact demonstrates this as a **computational toy claim**; it does not claim to reproduce human memory or biological synapses.

Problem and mechanism. Short-term memory requires information to remain usable without permanently changing long-term parameters. Dynamic synaptic state is one computational way to represent temporary information. In our educational substrate, a scalar weight represents the current strength of an association. A learning event uses the simplified rule **$w(t+1) = \delta w(t) + \eta \cdot pre \cdot post$**: δ preserves part of the old state, η controls the new write, and pre/post are active for the presented pair. With no new learning, the toy uses **w(t+n)=δ■w(t)**, so the state fades over time.

Evidence. Garcia Rodriguez, Guo & Moraitis (2022) introduced STP Neurons whose synapses have state propagated through time and evaluated short-term learning and forgetting across dynamic tasks [1]. Kozachkov et al. (2022) compared recurrent networks with and without STSP and found that both could maintain working memory, while STSP networks showed more brain-like activity and greater robustness to network degradation [2]. Chrysanthidis et al. (2025) modeled interactions between episodic memory and short-term recency using synaptic mechanisms operating across timescales [3]. These studies motivate the mechanism; they do not establish that our scalar rule is a complete biological theory.

Trade-off and artifact. The learner teaches *Apple* → *Red*, lets the Red trace decay, then teaches *Apple* → *Green*. Red and Green are stored as separate scalar traces in our implementation. Green does **not** erase Red. Instead, the traces compete at readout, where the toy selects the largest current weight. Higher persistence helps an older trace survive; stronger plasticity makes new information change the state more quickly. The final challenge lets the learner vary persistence/decay and delay and test a prediction.

Landscape. Static Transformer parameters remain fixed during inference unless another changing state or memory mechanism is added. Recurrent networks can store information in changing hidden activity. Short-term-plasticity models instead make synaptic variables part of the temporary state. STPN is a representative learned short-term-plasticity architecture [1], while Kozachkov et al. provide a working-memory comparison using RNNs with and without STSP [2]. Our artifact is not a benchmark against either system; it trades scale and biological detail for direct inspection and sub-minute reproducibility.

BDH and BDH-CQ. Kosowski et al. (2025) describe BDH (The Dragon Hatchling) as a brain-inspired language-model architecture whose working memory during inference relies on synaptic plasticity with Hebbian learning using spiking neurons [4]. Our toy isolates the intuition of a changing synaptic state and is **not** BDH. BDH-CQ (Engdahl et al., 2026) extends the BDH family with in-context learning and recurrent latent reasoning, using an evolving recurrent memory during inference [5]. Its role here is contextual rather than an implementation claim: our scalar toy does not reproduce BDH-CQ's latent reasoning mechanism.

Limitation. The most important limitation is the substrate itself: one scalar weight, simplified Hebbian learning, multiplicative decay, and argmax readout cannot capture the full dynamics of biological synapses or the high-dimensional state of frontier architectures. The correct takeaway is narrower and testable: **a changing synaptic state can serve as a temporary computational trace, and its persistence and new learning affect what a simple readout retrieves.**

Primary sources: [1] https://proceedings.mlr.press/v162/rodriguez22b.html [2] https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1010776 [3] https://www.nature.com/articles/s41598-025-12611-5 [4] https://arxiv.org/abs/2509.26507 [5] https://arxiv.org/abs/2608.09888